

Can Funding Fix Rural Physician Shortages? Evidence from a Location Choice Model

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Abstract

Physician geographic density affects access to care, especially in rural areas. This paper examines the location choice of early career physicians which includes standard demographics, practice environment proxies via Graduate Medical Education (GME) programs, and personal ties to the location. Using these demand parameters, I study the impacts of workforce policies by estimating the changes to rural probability when relevant attributes are varied in counterfactual analysis. I find that previous ties are major drivers of practice location: physicians have 49 times higher odds of choosing the location they did their residency in, but effects are halved in rural areas. Higher local hourly wages have little to no effects. Rural residency training exposure works through fostering previous ties to the location, underscoring the importance of local attributes in policy investments, and the potential of initiatives besides traditional approaches.

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1 Short Introduction

The geographic distribution of physicians represents one of the most persistent policy challenges in U.S. health care delivery. Despite \$2 billion in annual federal funding through Medicare, Medicaid, and targeted shortage initiatives ([HRSA, 2024a](#); [Congress, 2020](#)), concerns about the geographical distribution of the physician supply continue to worsen. Rural residency positions go unfilled at higher rates than urban positions despite their regulated and desirable nature ([Government Accountability Office, 2017](#)), and 73 percent of Health Professional Shortage Areas remain in shortage status for at least a decade after designation ([Markowski et al., 2023](#)).

This paper examines why current policies fail to resolve geographic physician shortages by analyzing the complete location choices of more than 58,000 US physicians completing training between 2016-2019. Using American Medical Association accreditation data linked with multiple public sources, I track physicians’ major life locations: birthplace, medical school, residency, and their first practice choice. This comprehensive trajectory data reveals that personal geographic ties dominate location decisions.

I make three contributions to the literature on physician distribution and health workforce policy. First, I document that personal ties—birthplace, medical school, and residency location—have significant effects relative to traditional levers such as financial and training exposure. Physicians are 49 times more likely to practice where they completed residency and 4 times more likely to choose locations matching their birthplace or medical school, while rural training exposure absent these ties increases rural practice probability by only 2 percentage points. Second, I show that Graduate Medical Education (GME) infrastructure serves as a powerful proxy for practice environment quality beyond its training function. Locations with more residency programs attract physicians even after training completion, suggesting GME creates professional networks and practice environments that enhance location appeal. Third, I develop a structural model enabling the simulation of outcomes from different policy designs, demonstrating that achieving even modest increases in rural physi-

cian supply through short-term investments require exponentially larger efforts than current programs provide.

The findings contributes to the analysis of persistent, selective shortages despite substantial federal investment. Without leveraging personal ties to locations, current incentives cannot overcome the disproportionate pull of urban markets. Rural training expansions attract few physicians if locations cannot produce practitioners with pre-existing geographic connections. This suggests that effective shortage policies must either develop long-term local pipelines starting before medical school or selectively target GME expansions to locations already producing medical students with regional ties. For locations unable to support either approach, alternative delivery models including telehealth and scope-of-practice reforms may be more cost-effective than continuing current physician recruitment strategies.

2 Full Introduction

The United States has spent roughly \$2 billion in historical and annual funding on physician shortage initiatives through Medicare, Medicaid, and targeted federal programs, yet efforts do not seem to fully address shortage concerns. By 2037, 60 percent of rural areas are projected to face physician shortages as the gap between supply and demand for physicians overall rose to above 200,000 personnel ([HRSA, 2024b](#)). 73 percent of designated Health Professional Shortage Areas remain underserved for over a decade after designation ([HRSA, 2024a](#); [Congress, 2020](#)). Rural populations, the majority of underserved locations, are aging faster than urban areas while simultaneously losing hospitals and physicians at accelerating rates. When combined with correlating poorer health in location with a thin supply of physicians, such observed uneven distribution raise concerns for substantial welfare losses: rural Americans travel two to three times as far for care, experience higher preventable mortality rates, and increasingly rely on emergency departments for primary care ([GAO, 2021](#)). Similar outcomes can exist outside rural locations as well, given underserved pockets exist in primarily urban areas and qualify for certain shortage status. Understanding why current policies have yet to address these issues despite substantial investment is crucial for designing effective interventions.

If personal geographic ties and professional networks drive location choices more powerfully than other incentives, especially for persistent shortage areas, then policies may require diversification and expansion in implementation, to accommodate the underlying determinants of physician distribution. While some prior work documents that physicians prefer urban locations, the specific attributes that make locations attractive remain understudied understood, namely professional networks, career opportunities, and practice infrastructure. Measuring the local practice environment’s quality beyond simple rural-urban classifications therefore is required in a framework with new information for health workforce and equity policy.

Prior research on physician location choice faces a few other critical empirical challenges.

First, most studies document physicians only from their entry into medical training, rather than their lifetime trajectory, although such information is highly desirable in consumer good demand models for individual heterogeneity (Nevo, 2001; Bayer et al., 2007). Estimates of rural training programs’ effectiveness without such consideration for personal ties cannot separate policy-induced improvement from potential pre-existing rural preferences, as physicians sort into locations based on sorting as well as potential policy impacts. Additionally, since most evaluations have been highly specific— focusing on retention statistics of certain locations or incentives— or literature reviews of a variety of programs, the methodological challenge is to analyze the entire population of physician and their response to different programs, rather than specific locations or incentive types.

This paper overcomes these challenges by constructing lifetime location histories for a sample of approx. 58,000 physicians completing training between 2016-2019, tracking their birthplace, medical school, residency, and first practice location. The model builds on choice modeling and a rich dataset to discuss policy via its influence on the geographical choice. This setup allows for flexible policy discussions via the effects on choice attributes. In addition to traditional policy channels, I also leverage certain degrees of random variations from the National Resident Matching Program and the historically rigid structure of GME programs to identify the effect of personal ties and GME programs on practice decisions. Counterfactuals capture the change to a specific outcome— rural practice probability— when policy changes different aspects. The results generalize to the impact that funding has on the location choice of physicians, assuming a pathway from health care dollar to attributes, then from attributes to the geographical distribution of physicians.

I model physician location choice using a random utility framework where each physician selects among state-rurality combinations to maximize utility. Utility depends on location attributes (compensation, climate, demographics), personal ties variables (or “home-coming” effects: birthplace, medical school, residency connections), and Graduate Medical Education infrastructure (program diversity, training positions). The choice set varies by specialty,

reflecting different labor market opportunities across specialties.

I estimate the conditional logit model by maximum likelihood, exploiting within-physician variation in location attributes, individual attributes and corresponding choice sets for identification. One key identifying assumption is that conditional on birthplace, medical school, and demographics, residency location is uncorrelated with unobserved practice preferences. This assumption leverages the National Resident Matching Program’s institutional constraints, which force over a quarter of physicians beyond their top choices, creating plausibly exogenous variation in training locations. It also utilizes the historical planning of GME programs to include this system as an elegant proxy for practice environment. Established as a nationwide subsidy system for physician training, GME programs vary dramatically across locations due to historical allocation patterns maintained since the 1997 Balanced Budget Act, potentially providing variations in practice environment quality that affects physician choices independently of rural-urban status.

To address potential endogeneity concerns, I implement several robustness checks to test for sensitivity to choice set definition, GME effects correlation with rurality (by sequentially adding controls and testing interaction effects), as well as various tests for the selection components in residency match coefficients. The stability of GME coefficients across specifications, ranging from 1.797 to 2.244 for program diversity, supports their interpretation as distinct infrastructure measures rather than rural-urban proxies.

The primary data source is the American Medical Association Masterfile, supplemented with GME program data from ACGME, specialty-specific compensation from the Bureau of Labor Statistics, and location characteristics from the Census Bureau. Processing of physician inputs allow for tracking of their birthplace, rural origin and foreign training. The dataset’s unique strength lies in capturing complete physician trajectories and the characteristics of the choice physicians make: from the involvement of their profile to that of the market structure and the choices they consider. Among the sample, only 1 percent completed rural residency training and 10 percent originated from rural states, highlighting the

pipeline challenge facing rural healthcare.

The results reveal that personal geographic ties dominate physician location decisions. Physicians are 49 times more likely to practice where they completed residency, 4.5 times more likely to choose their medical school location, and 4.3 times more likely to return to their birthplace. These effects overshadow traditional policy levers: rural residency training absent other geographic ties increases rural practice probability by only 2 percentage points, while a one standard deviation increase in compensation has negligible impact on rural choice probability. GME programs powerfully shapes location appeal, not just for their direct beneficiaries but also for full-fledged physicians. An additional residency program increases location choice probability, while additional positions within programs actually decrease attractiveness, suggesting physicians value professional diversity over training volume. Counterfactual simulations demonstrate that achieving even a 1 percent increase in rural physician share would require establishing 11 new residency programs in rural areas, nearly doubling the current rural GME infrastructure, or recruiting 11 times as many rural-origin medical students.

This paper contributes to two literatures. First, it extends research on physician labor markets by documenting that personal geographic ties and professional networks play a major roles in location decisions while simultaneously are emerging policy tools. I show that the combination of lifetime geographic ties explains more variation in U.S. physician distribution than all other factors combined. Although extensive literature has covered compensation and the supply of physicians, less attention has been drawn to the role of amenities and their components. As seems likely with any location choice, these factors are central to the utility function of physicians and essentially the success of workforce programs recruiting physicians. Therefore, modeling the choice of physicians in terms of amenities, compensation, and policy variables not only yields demand estimates of a specific population but also an evaluation of place-based policies.

Second, the paper advances understanding of how medical education infrastructure affects

healthcare delivery beyond its training role and provides insights into future expansions of the government-subsidized physician training programs. Consistent with past findings on the positive effects of Academic Centers, I establish that GME program diversity serves as a powerful signal of practice environment quality that attracts physicians independently of compensation or rurality. Government subsidies for physician supply can be described by a “fixed-cost” component that is GME, combined with a “variable-cost” of the place-based initiatives described above, which can vary based on funding policy. Recently, as evidenced by the Affordable Care Act’s Section 5503 and the Consolidated Appropriations Act (CAA) of 2021, even the “fixed-cost” GME system is earmarking positions towards shortage areas. This mechanism suggests GME expansion could address physician shortages not through training more physicians locally, but by creating professional environments that attract physicians from elsewhere, provided those physicians have pre-existing geographic connections to leverage.

3 Background and Literature Review

3.1 Physician Location Choice as Utility Maximization

This paper first contributes to the expansive research on physicians behavior in economics (Arrow, 1963; McGuire, 2000; Nicholson and Propper, 2011). The random utility maximization framework pioneered by (McFadden, 1974) provides the theoretical foundation for modeling choices, while Sims et al. (1982) provides a comprehensive analysis of such models in policy analysis. In this paper, a physician chooses the option that maximizes their utility, which depends on location-specific and individual attributes. Policy can enter in the outcome (geographical distribution), the attributes of location (place-based attributes as discussed in Cabral et al. (2025)), or the attributes of the physicians. Nicholson (2002) applies this setup to study physicians’ specialty choices and finds that medical students are highly responsive to expected income differences between specialties, which suggests increasing the relative

earnings in a specialty (like primary care) would lead more entrants to choose that specialty. [Agarwal \(2015\)](#) models a two-sided matching market. This study shows that residents are willing to sacrifice salary for desirable training programs, which keeps residency wages below the residents' marginal product, rather than the Match being an anti-competition device. Throughout this body of research, the consensus is that physicians weigh compensation along with training quality and other amenities when selecting positions. More recently, [Costa et al. \(2019\)](#) examines Brazilian physicians' practice location choices with a two-sided market, where the supply describe the combined physician activities in both the private and public sectors, a feature of the Brazil market. They show that physicians derive significant utility from working near their place of birth or where they trained. [Falcettoni \(2018\)](#) quantifies the impact of financial incentives on U.S. physician distribution. With incentives like loan forgiveness and salary bonuses, the appeal of location increases by 1.2 percent, proxied by the probability of choosing a rural practice.

3.2 Current Policies and Market Failures

The persistent shortage of physicians in certain areas raises the question of designing policy for market failures. [Newhouse et al. \(1982\)](#) discusses the policy justification for government interventions by questioning if the willingness to receive lower wages and urban preference are sufficient policy rationales, showing under assumption of diffusion when the marginal physician arrives in a location that many rural communities may simply lack the population and demand to support more physicians. Some sparsely populated areas would remain underserved no matter the subsidies, implying the distribution could already be economically optimal.

Policy interventions can be predicated on two market failures: (1) positive externalities of physician services (eg, social benefits from their services are not reflected in wages due to rural demographics and profitability) and (2) information asymmetries deterring physicians from rural opportunities. Assuming these market failures, financial incentives subsidize earnings

so supply can go up, or rural training opportunities provide the exposure that introduce people without information about underserved practice. In practice, the inconsistent efficacy of both financial incentives and underserved area training efforts suggests that the required wage premium to entice physicians to underserved areas is prohibitively high, or that other factors are at play beyond wages and information. Despite numerous federal incentive programs, physician shortages remain persistent. According to HRSA (2024), there were 7,501 designated primary care HPSAs across the U.S. (covering 75 million people), and two-thirds (66.5 percent) of these HPSAs are in rural areas. This underscores that much of rural America continues to face inadequate access, and it highlights the importance of distinguishing fixable barriers from fundamental demand constraints when designing policy.

In a similar vein, [Rosenthal et al. \(2005\)](#) finds that although the total number of physicians roughly doubled between 1979 and 1999, small and remote rural towns still lacked certain specialties. They distinguish between “selective” shortages in underserved yet potentially viable markets versus “persistent” shortages in areas that cannot sustain a physician economically- termed “pockets of shortage” in an earlier study by [Rosenblatt and Lishner \(1991\)](#). Indeed, recent evidence shows that many officially designated shortage areas remain chronically understaffed: [Markowski et al. \(2023\)](#) report that over 73 percent of Health Professional Shortage Areas (HPSAs) stayed in shortage status for at least a decade, with no significant changes to physician density and mortality outcomes.

Despite substantial investment, financial incentive programs may not resolve preference-based distribution universally. Multiple studies document negligible or short-lived impacts from these policies. For example, Medicare’s HPSA bonus payments (a 10 percent bonus for services in shortage areas) have had virtually no effect on where new physicians set up practice, rather even led to cutbacks where physicians treat fewer patients when examined under the McGuire-Pauly Synthesis of Supplier-Induced Demand model ([Brunt, 2025](#); ?). [Markowski et al. \(2023\)](#) finds no significant effects in terms of density and outcomes, while [Chou and Lo Sasso \(2009\)](#) finds that HPSA designations made underserved areas

more attractive only to physicians with no educational debt, indicating a clear stress on the approach when grounded in conventional means. Ironically, those with the highest debt (above \$100k) were least likely to choose an HPSA practice, despite loan-forgiveness programs targeting them. They concluded that the modest financial benefits of programs like the National Health Service Corps do not outweigh the non-monetary drawbacks of working in those areas. Similarly, loan repayment and scholarship programs often succeed in recruiting physicians for a few years, but retention after the service obligation ends is low: many participants leave rural practice once their commitment is fulfilled (implying the underlying location preference was unchanged).

3.3 Personal Ties as Dominant Location Determinants

Across many studies, personal geographic connections—such as one’s hometown and places of training—emerge as the strong predictors of where physicians choose to practice. [Wade et al. \(2007\)](#) examines a cohort of physicians in Indiana and found that those from non-metropolitan hometowns were 4.7 times more likely to practice in non-metropolitan areas compared to peers raised in metropolitan areas regardless of distance, while [Hughes et al. \(2005\)](#) finds the same tie using high school information. A more recent study by [Hu et al. \(2022\)](#) confirms that rural upbringing reliably predicts rural practice entry, even more so than social network proximity. [Brooks et al. \(2002\)](#) reviews different groups of studies ranging from pre-medical-school to residency to compare “nature” and “nurture” factors related to rural practice. They find that the pre-medical-school, early-in-life interventions are the most influential of all.

[Fagan et al. \(2015\)](#) find that among family medicine graduates, 55% practice within 100 miles of their residency location and 19% within just 5 miles, illustrating tight local capture for primary care specialties. This persistence varies by specialty, with family medicine (56%) and psychiatry (55%) showing stronger retention than surgical subspecialties like thoracic surgery (32%) and plastic surgery (31%) (AAMC 2023). These findings underlines the policy

logic that relocating or expanding residency slots serves as a direct lever for shifting the long-run geographic distribution of the physician workforce.

Figure 1 illustrates the most popular choice from each stage of life for physicians in the sample, exhibiting intense and persistent trends of staying in their birth location, which aligns with past literature findings.

[Figure 1 about here.]

Such strong “home-coming” tendencies have major policy implications for how straightforward and intuitive they are: physicians are more likely to come home, but the rural physicians are not common given the substantial investments required to become one. The dynamics create a self-reinforcing cycle in physician distribution. Areas without strong education investments are more likely to be without the infrastructure to train them, shrinking the future pipeline of providers with ties to those communities. In contrast, areas that already have robust medical communities tend to “replenish” themselves – local students go into medicine, and often return home to practice. Over time, this means pre-existing personal preferences and attachments perpetuate geographic imbalances. Policies that try to influence new physicians’ choices (through training or incentives) may face these deeply rooted preferences. Figure ?? compares the states at the lower quartiles college degree population and those with in the same quartiles with physician birth. The solution to such self-perpetuating issue may be to prioritize students with rural backgrounds and intent, i.e., “grow your own” (GYO) programs, similar to findings in (Costa et al., 2019).

3.4 GME Infrastructure as Practice Environment Proxy

Training opportunities is a major educational and informational incentive. In this paper, I use training opportunities to refer to either GME subsidized programs and positions (standard system) or place-based funded programs by rural or underserved basis such as Teaching Health Center Graduate Medical Education (THCGME). Originally created based on the

profound role that policy plays in ensuring a high-quality and equitable workforce, GME is a system of funding to reimburse physicians in training, organized by programs that are affiliated with teaching hospitals. Theoretically, GME-subsidized positions and the programs they belong to do not need to be urban: they are roughly by accreditation and funding¹ basis. In conjunction with the hospitals that contract the residents (physicians in training), GME pay for the portion of their salary corresponding with services completed for Medicare patients. GME’s contributions to this system is a “fixed cost” component that train physicians for the national supply, as opposed to any “variable cost” spent on rural/underserved place-based policies.

However, by its nature, GME current system is highly urban ([Adhikari et al., 2025](#); [Mullan et al., 2013](#)). Since the 1997 Balanced Budget Act, where these GME-subsidized programs and positions locate are not based on physician demands, but rather other institutional characteristics, such as the cap at the time when the program was established (similar to how the Per-Resident-Amount, the cost of training one resident, is established at the time when the system was initiated), the availability of academic physicians, the population of Medicare patients, etc. This structure is highly complex and with a certain degree of stickiness, at least to the extent that funding is involved, given hospitals can pay for residents without allotted subsidies as well. This long-static distribution of training slots means that the local supply of residency positions has strongly influenced physician workforce patterns, but does not adjust to physician preference, creating a degree of exogeneity.

There is an emerging movement to add more rural programs to tackle shortage, on top of general expansion of the supply. Once used with a rural/underserved qualifier, such as earmarked new positions that require shortage designation to qualify ([Congress, 2020](#)), GME starts to resemble the place-based, “variable cost” initiatives. The impacts of these new GME programs and positions, as well as any other “variable cost” initiatives are highly relevant to Medicare and Medicaid funding and the health workforce - two urgent and dynamic debates

¹For ACGME terminology on program and position, see [ACGME \(2024\)](#).

(GAO, 2025; HHS, 2025).

The availability of Graduate Medical Education (GME) infrastructure—residency programs and training positions in a location—also serves as a measurable proxy for the broader practice environment attractiveness of that region. Urban economics describes larger markets with agglomeration benefits such as knowledge spillovers, input-sharing, and better matching in labor markets (Duranton and Puga, 2003). In the healthcare context, teaching hospitals can elevate the quality of care not only within the institution but also at neighboring hospitals through practice-pattern diffusion and technology transfer, making the location choice more appealing and contributing to the utility function. Burke et al. (2023) look at the presence of an Academic Medical Center in a region and find that such centers produce spillover benefits – Medicare patients treated at non-teaching hospitals in high-AMC areas had significantly lower 30-day mortality than those in markets with no AMC. A strong teaching hospital can lift outcomes in surrounding facilities, likely via specialist availability, diffusion of innovations, and serving as a referral center.

Furthermore, the breadth of specialties and fellowships offered in a location signals to all physicians the professional resources available. Location choices with more specialty programs promise subspecialty backup, collegial learning, and career development opportunities that isolated rural practices typically cannot match.

4 Additional Background

4.1 Physician’s Role in Care Delivery and Outcomes

The geographic distribution of physicians fundamentally shapes healthcare delivery, as providers constitute the central component of care supply with direct impacts on mortality, quality, and costs. (Pines and Hollander, 2008) demonstrate that shortages of specialists increase mortality rates in emergency departments, where rapid access to specialized expertise can determine patient survival. This finding extends beyond emergency care: (Woodworth,

2020) shows how specialist shortages create congestion in life-saving care delivery, forcing delays in treatment that compound health risks. The mechanisms are straightforward: when specialists are scarce, patients face longer wait times, delayed diagnoses, and deferred procedures. These supply constraints bind most severely in rural and underserved areas, where the absence of even a single specialist can eliminate entire categories of care from local health systems.

Beyond availability, physician quality and practice patterns generate substantial variation in both health outcomes and healthcare costs, especially in rural areas. (Doyle et al., 2010) provides compelling evidence that physician quality directly translates into financial consequences through diagnostic ordering patterns, with lower-ranked physicians generating higher downstream costs through excessive testing. This relationship between physician characteristics and healthcare spending extends the literature on practice style variations (Grytten and Sørensen, 2003) The implications are particularly stark for mental health services, where (?) recently documents how provider shortages and quality variations lead to patient disengagement from follow-up care, creating cycles of acute crisis management rather than continuous treatment with impacts on long-term mortality. These quality dimensions interact with availability; areas with physician shortages often struggle to attract high-quality providers, thereby compounding access problems with quality concerns. (Chan et al., 2006) documents twice as long travel time for patient in rural locations to obtain care, while (Garcia, 2019) finds that preventable deaths are more prevalent in rural areas compared to urban counterparts.

Supply-side factors ultimately drive much of the observed variation in healthcare utilization and spending across regions. (Finkelstein et al., 2016) find that among Medicare beneficiaries, local provider density and healthcare facility availability account for more than half of geographic variations in care utilization. The concentration of providers not only affects the volume of care but also its composition, with specialist-dense areas showing different treatment patterns than those relying primarily on generalist care. U

4.2 Rural Shortage Policy

Federal efforts to address rural physician shortages have evolved through multiple policy approaches. The Health Resources and Services Administration (HRSA) has pioneered several initiatives that can be categorized into three main strategies: training expansion, financial incentives with service commitments, and capacity redistribution. As previously introduced, they could be conceptualized as a variable cost given their place-based nature.

The first approach, expanding training in rural settings, is implemented through programs such as Teaching Health Center Graduate Medical Education (THCGME) and Rural Residency Planning & Development (RRPD). THCGME provides direct and indirect GME payments for primary-care residencies based in community health centers, Rural Health Clinics, and critical-access hospitals, with positions allocated outside the Medicare GME cap. By 2022, THCGME supported more than 2,000 new primary care and dental practitioners who completed training and provided care for previously underserved patients (Department of Health and Human Services, Health Resources and Services Administration, Bureau of Health Workforce 2022). RRPD complements this by providing one-time planning grants to help hospitals secure ACGME accreditation for new rural residencies or rural tracks. In the same spirit of starting at training, this paper discusses the potential for increasing variety in local healthcare systems by establishing fewer rural hubs before delivering care to neighboring populations, in direct relation to THCGME and RRPD’s approach.

The second strategy employs financial incentives tied to service commitments. The National Health Service Corps (NHSC), established under the Emergency Health Professional Act, provides scholarships and tiered loan repayment in exchange for multi-year service in Health Professional Shortage Areas (HPSAs). In FY 2022, NHSC placed approximately 20,200 clinicians with 7,400 serving at rural sites. However, rural placements lag relative to rural statistics: only 32% of NHSC physicians serve in rural areas despite rural regions comprising 67% of HPSAs (U.S. Department of Health & Human Services, Health Resources and Services Administration, Bureau of Health Workforce 2024). Holme (2005) finds that

NHSC accounts for only a modest 10% rural retention after accounting for heterogeneous preferences, consistent with this paper’s finding of minor effects from rural training once self-selection and rural unobserved preference are considered.

Of the three major approaches discussed so far (financial incentives, establishing new centers, increasing current cap), the Affordable Care Act’s Section 5503, and the GME provisions in the Consolidated Appropriations Act (CAA) of 2021 or the “cap raise” policy, are the most recent relevant policy to this analysis. Under the Affordable Care Act’s Section 5503, unfilled positions in primary care and rural areas are reclaimed and redistributed. These slots are then allocated to hospitals with smaller programs in larger states, those in rural counties, or the ten states with the highest primary-care HPSA-to-population ratios. [McNamara and Pineda-Torres \(2025\)](#) find that this initiative has increased teaching intensity by around nine percentage points, and recruitment into high need primary care programs by 23 %. Similarly, the CAA of 2021 created 1,000 new Medicare-funded slots (up to 200 per year for five years starting July 1 2023). Although this holistic approach acknowledges both the overall need to expand the supply of providers and geographic/specialty-specific shortages, its rural care expansion faces the same demand issues that created the unfilled slots in the first place: urban competition and low take-up of rural training. Figure [14.3](#) reiterates the issue in visualizing the different nature of rural and urban response to new slots: rural programs struggle to retain, even when more training slots are redistributed.

[Figure 2 about here.]

The central empirical question is whether slot creation (RRPD, THCGME) and debt relief (NHSC) are jointly sufficient to offset the powerful centripetal pull of urban labor markets, especially given their smaller scale and funding duration relative to GME, which is essentially incorporated into Medicare and Medicaid. As demonstrated in this study, a strong urban preference may also hinder retention and recruitment in programs located near rural areas. This paper addresses these issues by proposing a common ground for rural care delivery based on physician preference and the implementation of GME funding.

Despite the rigidity of the GME cap, which is also being reformed, as evident by the aforementioned provisions, its official role and deep ties to the top of the pipeline at training stages present the next potential addition to ongoing initiatives.

Overall, ongoing and past approaches show meaningful but incomplete gains. Markowski et al. (2023) find that Health Professional Shortage Areas (HPSA) designation, which includes both financial incentives and foreign medical personnel (IMGs) recruitment, has no significant impact on physician density or mortality outcome after a decade. The same study highlights the substantial barrier in professional isolation, which this paper identifies through physicians' preference for a diverse and established local healthcare infrastructure. To overcome this barrier, I suggest a hub-and-spoke (or train-and-ship) approach as a middle ground between training and rural care. Timely reforms in care delivery, most notably telehealth, have created the space for new changes in policy implementation that carry on the mission of HRSA's initiatives.

4.3 Overview of GME's Role in Physician Training

GME funding operates through a complex reimbursement system where federal dollars subsidize resident salaries while hospitals serve as the formal employers, creating a unique labor market where physicians effectively pay for training through reduced wages. Resident physicians' salaries are a function of determined parameters (e.g., the Per Resident Amount or PRA, established since the first GME positions were federally supported in the 1960s), inflation adjustments, specialty, and cost-of-living considerations. Both the sponsor of the residency program (the overall administrator of the internship) and GME pay the resident during residency.

Federal funding reimburses and subsidizes part of physicians' salaries, while hospitals (or ACGME training sites, which can be clinics or other institutions) are the residents' formal employer. Hospitals pay the rest of physicians' salaries and benefits after federal subsidies (Heisler et al., 2018). In discussing potential antitrust issues in the centralized procedure

that people get assigned into their residency, (Agarwal, 2015) discusses the willingness of physicians to pay for their training in the form of accepting a lower salary during residency, which makes GME dollars even more impactful relative to higher salaries. (Falcettoni, 2018) confirms this heightened impact via the loan forgiveness and salary incentives on physician location.

Although GME is by definition active at the residency stage of the physician supply, the eventual distribution after training and commitments to patient communities is what policies are after, not short-term retention. Realistically, physicians who have finished training do not share the same stakes as those entering training, so federally funded policies do not have the same reach to this population if they remain within training programs. Nevertheless, we want to retain physicians after they have completed training to address any possible concerns about shortages.

5 Data

5.1 Physicians

The main dataset cover most US physicians who completed their *final* stage of medical training, residency or fellowship, between 2016 and 2019 from the American Medical Association (AMA) Masterfile (American Medical Association, nd). The time period aims to predate the COVID pandemic to avoid shocks to labor trends, while capturing the choices of physicians that are as close to 2021, when (Congress, 2020) introduced the residency cap raise as one of the motivations, as possible². The specialties selected include both primary care and surgery, except for subspecialties that BLS data do not explicitly include in their occupation codes; 7 specialty groups based on the Standard Occupational Classification are included. Table 1 reports summary statistics for the physicians in the sample, followed by the specialty

²A robustness test is conducted which shows no practical changes to the key coefficients when the model limits to the first 2 years or the last 2 years of the study period (Table 12).

summary in Table 2.

[Table 1 about here.]

[Table 2 about here.]

The primary source for physician trajectory is the American Medical Association (AMA) Masterfile, supplemented with residency–site records from the Accreditation Council for Graduate Medical Education (ACGME). The choice unit is recorded exactly one year after training for computational size as well as generality; each physician therefore appears once. After the standard cleaning, the analytical file contains around 58,000 physicians spread across seven broad specialty groups as defined by the SOC system and 96 units at state–rural level (Table 2.

5.2 Location Attributes

The observed chosen choice for each physician is the unit with state and rurality attribute of their location one year after medical training. Each choice unit is a combination of (state, rurality)³, following the practice in (Costa et al., 2019) to retain the rurality of each location choice while keeping the choice set computationally compact. The choice set varies with specialty and year, as presented by Table ?? and discussed in Technical Notes.

[Table 3 about here.]

Table 4 presents the summary statistics of choice units in the sample. Rurality code is maintained by crosswalking back to the Core-Based Statistical Area (CBSA) code and Rural–Urban Continuum Code (RUCC). Physician compensation is proxied by state–specialty hourly wages from the Occupational Employment Statistics Model-Based 3-Year (OEWS-MB3) estimates constructed by Dey et al. (2019), lagged one year to avoid simultaneity and

³Physicians trained in Puerto Rico and U.S. territories are excluded because wage data are unavailable. Some states and DC are exclusively urban (NJ, RI, DC)

simulate physicians considering past salaries. To overcome missing data, I use several stages of imputing using timeline, state, and then regression-based imputation within the specialty. Figure 14.3 shows the distribution of salary by specialty group in the sample.

[Table 4 about here.]

[Figure 3 about here.]

The salary ranges in Figure 14.3 align with specialty-specific levels reported in past studies on physician earnings (Nicholson, 2002; Gottlieb et al., 2025), but with significant differences from other paper due to using employer-reported survey data of *physicians in the choice location* rather than W-2 income of the physicians examined as in Gottlieb et al. (2025).

5.3 GME programs and positions

The rural-urban disparity is mirrored in GME components. Table 5 displays GME data on the choice units included in the sample. There are clear differences in how GME funding is distributed and organized, which correlate closely with the scale and sophistication of the local health delivery system. GME positions, programs, and sponsors per unit come from the Area Health Resources Files; climate (mean minimum temperature), median household income, and racial composition come from the Census. All covariates are measured in the calendar year preceding the stated location.

[Table 5 about here.]

6 Model

6.1 Physician Location Choice Framework

I model physicians' location choices using a conditional logit framework following McFadden (1974). Each physician i in specialty s and year t chooses among a set of location alternatives

$\mathcal{C}_{st}$ to maximize utility. The utility that physician i derives from choosing location j is given by:

$$U_{ijt} = V_{ijt} + \epsilon_{ijt} = \mathbf{x}'_{jt}\boldsymbol{\beta} + \mathbf{z}_i \cdot \mathbf{x}'_{jt}\boldsymbol{\gamma} + \mathbf{m}'_{ij}\boldsymbol{\delta} + \epsilon_{ijt} \quad (1)$$

where V_{ijt} represents the deterministic component of utility, $\mathbf{x}_{jt}$ is a vector of location-specific attributes, $\mathbf{z}_i$ contains physician-specific characteristics, and ϵ_{ijt} is an i.i.d. Type I extreme value error term following McFadden's setup.

Each physician is indexed by their location records (birth, medical school, and residency), domestic birth status, foreign medical school (IMG) status, rural residency status, gender, degree type (DO/MD), specialty, and age. Each location is characterized by its rurality, a set of socio-economic descriptors (median household income, black population, bachelor's degree attainment), and a minimum temperature variable to proxy for weather conditions.

The choice set $\mathcal{C}_{st}$ varies by specialty and year, reflecting the fact that physicians in different specialties face different relevant labor markets (further information is available in Technical Notes). This approach ensures that each physician faces a realistic set of alternatives and accounts for the differences across specialties.

Key location attributes in $\mathbf{x}_{jt}$ include:

1. Rural _{j} : Binary indicator for rural locations
2. log(GME Funding _{jt}): Log of total Graduate Medical Education funding (in thousands)
3. GME Programs _{jt} : Number of ACGME-accredited residency programs per choice
4. GME Positions _{jt} : Number of residency positions per choice
5. log(Compensation _{jt}): Log of physician compensation measured in BLS compensation data for the specialty group and year
6. Additional location characteristics including median household income, temperature, and demographic composition, primary care share of GME positions

The physician characteristics $\mathbf{z}_i$ include indicators for foreign medical school graduation, rural residency training, age categories, and gender. I interact these characteristics with location attributes to capture heterogeneous preferences. For example, the interaction between foreign medical school graduation and rural location captures differential preferences for rural practice among international medical graduates.

The match-specific variables $\mathbf{m}_{ij}$ capture whether location j matches physician i 's home state (birth state), medical school state, or residency state. These variables account for the strong tendency of physicians to practice near their training or origin locations.

6.2 The Individual Choice Set

The construction of physician choice sets requires careful consideration of market feasibility constraints. Following McFadden (1974), I define feasible choices as locations where practice opportunities exist, either through employment positions or potential market entry. This definition encompasses both institutional employment (hospitals, clinics) and entrepreneurial opportunities (private practice). In practice, the choice set is the union of all unique choices across the study period of each specialty. This treatment of the choice set is straight-forward and intuitive since the alternative more sophisticated approach using employment data could suffer from capturing equilibrium outcomes rather than opportunity sets (Petrin and Train 2010). In other words, rather than using employment count, I use historically observed choice sets by specialty to avoid assuming a choice is not considered simply because it was not chosen as shown in employment data. This approach is particularly fitting in this study with unfilled positions in certain geographic areas.

The use of the entire study period leverages panel data richness to accurately reflect time-invariant structures of regulatory and fixed cost constraints. In other words, a feasible choice in one period is unlikely to become entirely infeasible in other periods since facilities and population needs are established, while a choice never filled in 4 years is likely truly not in the choice set due to nonexistent positions.

6.3 Estimation

Under the assumption that ϵ_{ijt} follows an i.i.d. Type I extreme value distribution (IEV), the probability of an individual i choosing alternative j becomes the probability that the difference between two Type I EV errors is less than to the difference between the deterministic portion of the chosen good and any other considered goods. Specifically, physician i chooses location j if and only if:

$$U_{ijt} > U_{ikt} \quad \forall k \in \mathcal{C}_{st} \quad (2)$$

This implies:

$$V_{ijt} + \epsilon_{ijt} > V_{ikt} + \epsilon_{ikt} \quad \forall k \in \mathcal{C}_{st} \quad (3)$$

The researcher does not observe the error, but the agent does, which represents any unobserved factors affecting the choice made. This error term is assumed to be random. Given the distributional assumptions, the probability that individual i chooses an alternative j can be expressed conditionally on the attributes of goods $j = 0, \dots, J$ in her choice set:

$$Pr(Y_i = j | X_{i0}, \dots, X_{iJ}) = \frac{\exp(V_{ijt})}{\sum_{k \in \mathcal{C}_{st}} \exp(V_{ikt})} \quad (4)$$

I estimate the parameter vectors β, γ by maximum likelihood. The log-likelihood function is:

$$\mathcal{L}(\beta, \gamma) = \sum_{i=1}^N \sum_{j \in \mathcal{C}_{st}} y_{ij} \log(P_{ijt}) \quad (5)$$

where $y_{ij} = 1$ if physician i chooses location j and 0 otherwise.

Substituting the probability expression in (4) and expanding the deterministic utility as a function of attributes:

$$\mathcal{L} = \sum_{i=1}^N \sum_{j \in \mathcal{C}_{st}} y_{ij} \left[\mathbf{x}'_{jt} \boldsymbol{\beta} + \mathbf{z}_i \cdot \mathbf{x}'_{jt} \boldsymbol{\gamma} - \log \left(\sum_{k \in \mathcal{C}_{st}} \exp(V_{ikt}) \right) \right] \quad (6)$$

The first-order conditions for the location attribute parameters $\boldsymbol{\beta}$ then simplify to:

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\beta}} = \sum_{i=1}^N \sum_{j \in \mathcal{C}_{st}} (y_{ij} - P_{ijt}) \mathbf{x}_{jt} = \mathbf{0} \quad (7)$$

Where the contribution to the log likelihood from each individual is summed over the sample. Similarly, for the interaction parameters $\boldsymbol{\gamma}$:

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\gamma}} = \sum_{i=1}^N \sum_{j \in \mathcal{C}_{st}} (y_{ij} - P_{ijt}) (\mathbf{z}_i \cdot \mathbf{x}_{jt}) = \mathbf{0} \quad (8)$$

These first-order conditions show that at the optimum, the weighted sum of characteristics across all alternatives (weighted by the residual $y_{ij} - P_{ijt}$) equals zero. This is solved iteratively in Python as the probabilities P_{ijt} are nonlinear functions of the parameters.

All continuous variables are standardized to have mean zero and unit variance to facilitate interpretation of coefficients and improve numerical stability during optimization. The conditional logit structure effectively controls for physician-specific unobserved heterogeneity that is constant across alternatives, as these factors cancel out in the probability ratio.

6.4 Identification

The parameters are identified through variation in location attributes across alternatives within each physician's choice set and across choice sets. Formally, identification requires that the Fisher information matrix:

$$\mathbf{H} = \mathbb{E} \left[\sum_{j \in \mathcal{C}_{st}} P_{ijt} (1 - P_{ijt}) \mathbf{w}_{ij} \mathbf{w}'_{ij} \right] \quad (9)$$

be of full rank, where $\mathbf{w}_{ij} = [\mathbf{x}'_{jt}, (\mathbf{z}_i \cdot \mathbf{x}_{jt})', \mathbf{m}'_{ij}]'$ is the full vector of covariates.

The GME policy variables exhibit substantial geographic variation, with rural areas typ-

ically having fewer programs and positions but potentially higher per-position funding. This variation exists even within states, providing identifying variation after controlling for state-level preferences through the match variables.

The parameters γ (interactions) are identified through physicians with different characteristics facing the same choice sets but making different choices, while β (main effects) are identified through within-physician variation across the attributes of alternatives in their choice set.

7 Results

7.1 Descriptive Analysis on Geographic Ties

Among 58,646 physicians in the sample, 19.8 percent choose to practice in their birth state despite birth states representing only 2 percent of available choices. To further look into ties, I run a compact version of the main analysis to examine birth ties in specific:

$$\begin{aligned}
U_{ij} = & \beta_1 \cdot \text{birth_match}_{ij} + \beta_2 \cdot (\text{birth_match}_{ij} \times \text{option_popularity}_j) \\
& + \beta_3 \cdot \text{res_match}_{ij} + \beta_4 \cdot \text{med_match}_{ij} \\
& + \beta_5 \cdot \text{median_hhinc_std}_j + \beta_6 \cdot \text{log_comp_std}_j \\
& + \beta_7 \cdot \text{rural}_j + \beta_8 \cdot \text{meanmintemp_std}_j \\
& + \beta_9 \cdot \text{pc_blackpop_std}_j + \beta_{10} \cdot \text{primary_care_std}_j \\
& + \beta_{11} \cdot \text{gme_progs_std}_j + \beta_{12} \cdot \text{gme_pos_std}_j + \epsilon_{ij}
\end{aligned} \tag{10}$$

Table 6 describes the findings while Figure 2 graph the marginal effects with popularity of the location choice on the horizontal axis. Physicians who practice in their birth state versus those who leave. Homecomers are younger, more likely to be female, and far less likely to be foreign medical graduates. The 0.615 correlation indicates that locations with more "native" physicians in the potential pool tend to be chosen more frequently overall.

However, larger states both produce more physicians and attract more physicians.

[Table 6 about here.]

To examine whether birth ties compensate for location unattractiveness, I then estimate marginal effects across the continuous distribution of location popularity. Figure 2 shows that birth state match increases choice probability by 0.305 at the 10th percentile of popularity, declining monotonically to 0.276 at the 90th percentile: a 9.5 percent reduction. This pattern suggests personal ties partially offset disadvantages in less popular locations more than in highly popular markets.

[Figure 4 about here.]

These patterns validate the modeling choice to distinguish rural and urban markets within states rather than treating states as unified labor markets. The 96 percent exact-return rate among in-state practitioners descriptive suggests that residency location does have a strong pull, but the effect of training is local, meaning a rurally trained physicians are “attracted” to where they trained, rather than any other similarly rural locations. This nuance implies that policy is not changing physician’s mind about rural practice in general, rather about the specific location they trained.

7.2 Main Findings

The conditional logit choice model results are presented in Table 7.

[Table 7 about here.]

Physicians strongly prefer urban areas, they are twice as likely to choose an urban choice compared to a rural choice. The odds ratio is 0.479 for rural choices, calculated as $e^{-0.735}$, indicating that rural locations are chosen 52% less frequently than urban locations when all other factors are held constant. They also prefer choices with higher median income and

milder climate. Specifically, a one standard deviation increase in median household income increases the odds of choosing a location by 17.8% (OR=1.178), while a one standard deviation increase in mean minimum temperature increases choice odds by 26.1% (OR=1.261). The expected signs across specifications support the model specifications, including lower share of GME investment in primary care tendency to lean towards specialization and away from primary care. The primary care share coefficient of -0.023 translates to an odds ratio of 0.977, suggesting locations with higher primary care focus are slightly less attractive to the average physician in this sample. However, they do not show a lot of response to the current compensation of their specialty in the location. The log compensation coefficient of 0.012 is statistically significant but economically small, with a standard deviation increase in compensation increasing choice odds by only 1.2%. This may be due to the nature of this compensation variable being a measure of that specialty average compensation, rather than actual salaries that physicians get offered.

Demographic variables enter the utility function through interactions with the rural dummy variables. Physicians trained in foreign medical schools are significantly less likely to practice in rural locations, with the interaction coefficient of -0.486 reducing their already low rural preference by an additional 38.5% (OR=0.615), a finding that aligns with recent updates to the IMGs role in filling in shortage positions discussed in ([Hailat et al., 2025](#)). Also consistent with past literature, those who completed their residency in rural programs are more likely to work in a rural setting, with the interaction coefficient of 0.207 increasing rural choice odds by 23% (OR=1.230). Young physicians (under age 32) show a similar positive interaction with rural locations (coefficient=0.287, OR=1.333), suggesting early-career physicians may be more willing to practice in underserved areas.

A location that matches a physician’s residency location is 47.3 times more likely to be chosen than an otherwise identical location that doesn’t match. Although the interaction between rural training and rural choice aligns with past literature findings that rural incentives do have some effects in retaining physicians, the relative scale of the negative rural

coefficient and positive rural training exposure suggests the overall bias for urban areas is difficult to overcome via exposure alone. Specifically, the base rural penalty (OR=0.479) is only partially offset by rural residency training (OR=1.230), resulting in a net odds ratio of 0.589 for rural-trained physicians choosing rural locations—still 41% less likely than choosing urban locations.

Geographic ties are represented via dummy variables that are set to 1 for choices that match the physician’s location profile. These ties prove extremely important in decision-making, with residency having the largest effect. The birth state match increases choice odds by 4.249 times, while medical school state match increases odds by 5.410 times.

Finally, the training infrastructure attributes enter via the GME-funded program and position counts. Program variety seems to be the driver of appeal, while larger capacity does not attract physicians. Around the mean number of programs per choice, a marginal program increases choice odds by approximately 0.72% (assuming constant effects) since each additional 254 GME programs (one standard deviations) per unit increases the odds of choosing that location by 6.213 times, while each additional 1 standard deviation from the mean of position per unit actually decreases attractiveness with an odds ratio of 0.212. This combined with the control for GME primary care FTEs provide an sight on GME planning: at the same proportion of primary care to non-primary care residents in the cohort, physicians still prefer the location with more programs. The appeal therefore could be explained as a potential preference for practice location, rather than urbanity alone, since urbanity would not warrant *more* programs, but always warrants *bigger programs*, while a location choice with a diverse system of programs across the rurality spectrum is not always the most urban.

GME programs are accredited, specialty-specific, and administered by GME sponsors (ACGME). Therefore, the number of GME programs is directly related to the variety of specialties and the established infrastructure for teaching each specialty. In the context of this market, GME funding is therefore also establishing the expert network of teaching physicians and infrastructure for an active, sustainable environment. In other words, beyond

the immediate pull by having residency positions available, what matters to the physician supply that comes out of government funding is still the care-delivering institutions in which the programs are based, rather than the programs themselves.

8 Counterfactual Analysis of GME Implementation

Table 9 presents the results of counterfactuals based on the model. These probability changes translate into different cost-effectiveness ratios when considering implementation expenses. Table 10 compares the cost per additional rural physician across interventions.

[Table 8 about here.]

[Table 9 about here.]

[Table 10 about here.]

9 Robustness Checks

9.1 Residency Tie vs. Selection

The model assumes residency location is exogenous conditional on controls: if physicians select residencies where they intend to practice, β_2 in equation (1) may be biased upward due to this effect. I exploit the different institutional constraints on residency choice between two subgroups of physicians to test for this endogeneity issue. Foreign medical graduates (FMGs) face more limited residency selection through various visa requirements and more competition (NRMP, 2022). Under selection bias, we would expect $\beta_2^{\text{US}} > \beta_2^{\text{FMG}}$ since US graduates may have more liberty to choose their location, despite not with perfect randomness. In other words, β_2^{US} contains the extra selection bias and advantage that correlate with the choice on the LHS, while that of the FMGs can be assumed to have less variation based on selection.

Assuming additive form and homogenous effects in a simplified model, I run:

$$U_{ijt} = \beta_1 \mathbb{1}\{\text{Birth}_i = j\} + \beta_2 \mathbb{1}\{\text{Residency}_i = j\} + \beta_3 \mathbb{1}\{\text{MedSchool}_i = j\} + \mathbf{X}'_{jt}\gamma + \epsilon_{ijt}. \quad (11)$$

where the concern is that β_2 may be biased if $\text{Cov}(\mathbb{1}(\text{Residency}_i = j), \epsilon_{ijt}) \neq 0$.

Table 11 shows the opposite: FMGs exhibit a larger coefficient (4.030) than US graduates (3.760), a difference of 0.270 ($p < 0.001$). This pattern rejects the hypotheses that geographic ties don't matter and shows the difference in their effects across two groups with different levels of constraints in choosing their preferred residency location. The pattern is inconsistent with selection bias inflating the residency coefficient, though it does not prove exogeneity. The same test was run on foreign-born physicians (results are not included) where the coefficient is also larger than that of US graduates.

As a complementary test, I exploit variations in match competitiveness across specialties. Surgical specialties have lower match rates, forcing more physicians into non-preferred programs. In contrast, primary care specialties with more positions provide more room for selection into residency, making this impact more likely to be endogenous, yet they seem to consider the residency impact less than their colleagues in surgical specialties. Table 11 shows an effect of 3.947 for surgical versus 3.182 for non-surgical specialties. Consistent with the previous finding that selection doesn't fully drive the variations in residency match, this shows that those less likely to have done a residency they prefer since they are matching into surgical specialties show a stronger response via residency match.

Although these results do not prove exogeneity, they provide concrete evidence that variations from residency match are not purely driven by selection, using the different groups of physicians in the sample.

9.2 GME Programs and Rurality

The identification relies on the fact that the GME variables come from policies historically designed for training, and were only very recently (after the start of the study period in 2016) employed directly for adding new slots to shortage areas. Due to centralized regulations and high dependency on local infrastructure, these GME variables do not inherently address shortages. However, there is a concern that rurality accounts for any impacts these variables have on outcome, which would challenge the assumption that training variables are primarily a proxy for local health care facilities and professional networks, which are at the core of the counterfactuals.

The distribution of Graduate Medical Education infrastructure reflects historical allocation patterns rather than endogenous responses to physician choices. Table 5 shows stark disparities: urban locations average 228.3 residency programs compared to 1.6 in rural areas ($t=25.60$, $p < 0.001$), and 2,755 positions versus 17.5 positions respectively ($t=23.72$, $p < 0.001$). Notably, 100 percent of urban locations in the sample have some GME funding, while only 70.8 percent of rural locations have any GME presence. This ratio predates the study period and reflects strong consistence with funding caps that essentially froze GME distributions, the complex updates of which system is described in [Chen et al. \(2013\)](#).

The stability of GME effects also supports reasonable degree of independence from the rurality. Table 12 presents regressions progressively adding controls. The rural coefficient change from -2.458 independently (Model 1) to -1.489 when GME variables are added (Model 2), then stabilizes between -1.477 and -1.627 as additional controls enter (Models 3-5). Meanwhile, GME program effects remain remarkably consistent, ranging from 2.244 to 1.797 across specifications, as do position effects (-1.659 to -1.522). This stability suggests GME variables capture distinct infrastructure quality rather than merely proxying for rurality.

[Table 11 about here.]

[Table 12 about here.]

Adding GME variables to a rural-only model improves log-likelihood by 21,128 points, while adding all other controls (economic, demographic, and geographic ties) provides an additional 96,635-point improvement. The interaction test between GME variables and rurality is significant and but the main effects persist, indicating that while GME impacts differ by rural status, they represent more than simple urban-rural distinctions. The model approximates physician choices better when it knows not just whether a location is rural, but also its GME infrastructure. This suggests physicians care about training programs and academic medical environments independently of rural/urban status. These patterns support interpreting GME infrastructure as a predetermined measure of local medical practice environment quality rather than an endogenous policy response.

9.3 Choice Set Sensitivity

The baseline specification includes only locations historically chosen within each specialty-year. Table 12 tests alternative definitions of choice set on the rurality and residency match key variables: (1) baseline historically chosen locations, and (2) any location chosen by any physician. The residency coefficient ranges from 3.836 to 4.213—less than 10% variation. Birth and medical school coefficients show similar stability.

9.4 Temporal Stability

Geographic tie coefficients remain stable: birth match changes from 1.473 to 1.464. Testing for complementarities among multiple ties reveals negative interactions (-1.671, $p < 0.001$), suggesting diminishing rather than multiplicative returns.

Temporal stability checks show the residency coefficient is 3.830 for 2016-2017 and 3.883 for 2018-2019, a 1.4% difference. All key coefficients exhibit similar stability across periods.

Table 12 summarizes results across all tests. The residency coefficient ranges from 3.448 to 4.030, with coefficient of variation of 7.8%. The main specification’s estimate falls near the center of this range. Notably, tests designed to reduce selection bias yield higher coefficients,

reinforcing that main estimates are conservative.

10 Policy Implications and Welfare Analysis

10.1 Cost-Effectiveness of Current and Alternative Policies— NUMBERS NOT FINAL

10.1.1 Via GME

Although the CMS does not publish data on payments, using estimates of spending and supported FTEs from Villagrana (2024), the cost per resident position per year is on average \$91,000 per physician. The magnitude of geographic tie effects relative to policy instruments explains the limited success of current physician distribution programs. Table 9 demonstrates that adding 10 GME programs to every rural location increases rural practice probability by only 0.42 percentage points, from 5.96% to 6.38%. Given that establishing a new GME program costs approximately \$150,000 per position annually, and assuming 5 positions per program, this intervention would require \$[] million per state, or \$18.75 billion nationally, to achieve less than a one percentage point increase in rural physician supply.

What these findings suggest show that GME expansion can work as workforce policy lever, but are expensive and strictly regulated. One implementation under such constraints could be to invest via GME only in locations not entirely rural (so that the population can support GME programs, fiscally and all the above factors) but also not well-established. Figure 3 examines a specific case where smaller programs are joined together within 1 hour driving distance, together with any unfilled positions nearby, to move GME investments to a nearby current programs capable of supporting an expanded programs, harnessing the effects of more programs without the limitations of rural infrastructure.

[Figure 5 about here.]

10.1.2 Via Personal Ties

The dominance of geographic ties suggests fundamentally different approaches to physician distribution. Rather than attempting to overcome preferences with marginal incentives, policies could leverage existing ties through targeted recruitment and training location.

Increasing enrollment of rural-origin students by 25% would require identifying approximately 2,000 additional qualified rural applicants annually. Given that relatively rural students comprise only 10% of the sample despite representing 15% of the population, substantial untapped potential exists. The challenge lies not in capacity but in preparation—rural students systematically underperform on MCATs and have lower undergraduate GPAs, reflecting educational disadvantages rather than ability.

Pipeline programs beginning in secondary education could address these preparatory gaps at relatively low cost. A comprehensive program costing \$10,000 per student annually from grades 9-16 (high school through post-baccalaureate) would total \$80,000 per future rural physician—less than one year of NHSC support and far below GME expansion costs. Even if only 20% of participants ultimately practice rurally, the cost per rural physician remains competitive with current programs.

The counterintuitive finding that restricting choice sets to home states increases rural practice by 8.13 percentage points suggests scope for regulatory approaches. While mandatory service requirements likely violate constitutional protections, voluntary programs offering enhanced benefits for geographically restricted practice could achieve similar effects. For instance, an enhanced loan forgiveness program requiring practice within one's home state for ten years might attract physicians with strong geographic ties while screening out those likely to relocate.

11 Limitations

This analysis faces certain limitations. First, I observe only the first practice location, not long-term retention. Physicians induced to practice rurally through financial incentives may relocate after fulfilling obligations, while those with geographic ties may exhibit greater persistence. Without longitudinal data, I cannot distinguish temporary from permanent location decisions.

Second, I do not have actual policies in the model, and cannot separate causal effect of residency training from selection on unobservables that vary within Match constraints. While foreign medical graduates and surgical specialties provide variation in choice constraints, these groups may differ on unobserved dimensions that affect location preferences. The true causal effect of residency location might be smaller than estimated, though the robustness across subsamples suggests the qualitative findings hold.

Third, the dataset constructed require processing of data from various sources where there are a margins of error, either by text-processing or the nature of the dataset. AMA data have been discussed with certain overstated sample size or data not up-to-date, and OEWS data on compensation requires imputation for missingness, especially in smaller markets.

Finally, the model does not incorporate general equilibrium effects. Large-scale reallocation of physicians would affect wages, working conditions, and potentially the geographic distribution of patients themselves. The partial equilibrium estimates likely overstate the physicians that could be relocated through any single policy while understating the costs of comprehensive reform.

12 Conclusion

Geographic ties dominate economic incentives in physician location choice, with residency location increasing practice probability 49-fold compared to marginal effects of financial incentives. This finding explains both the persistence of geographic maldistribution despite

substantial policy intervention and the limited success of current programs. We operate at the margins of a distribution fundamentally determined by where physicians originate and train when we offer later-in-life interventions.

The welfare implications extend beyond rural health. Geographic clustering of physicians reinforces regional inequalities in health outcomes, economic opportunity, and human capital development. Rural areas lose not only healthcare access but also the economic multiplier effects of high-skilled professionals and the role models that might inspire the next generation to enter medicine. Urban areas, conversely, experience diminishing returns to physician density while bearing the costs of medical education for physicians who provide limited social benefit in already-served markets.

The ideal world policy suggestions would be to leverage geographic ties through targeted recruitment from underserved areas, or improve appeal through strategic placement of GME training programs. The former approach requires long-term investments in rural educational pipelines; the latter requires substantial capital investments in selective GME infrastructure. Both strategies imply decades-long commitments and acknowledgment that current physician shortages reflect structural factors rather than issues correctable through marginal incentives.

The modest effects of financial incentives do not imply physicians are unresponsive to economic factors, but rather that the economic rents required to overcome geographic preferences exceed societal willingness to pay. A rural premium sufficient to achieve geographic balance might require doubling or tripling rural physician compensation—transfers that would be both fiscally prohibitive and ethically questionable given rural-urban income disparities in the general population.

Future research objectives include examining dynamic aspects of location choice, including retention, spillovers to other health professionals, and the production function for rural health. For example, understanding whether nurse practitioners and physician assistants exhibit similar geographic ties could inform whether expanding scope of practice laws offers a more feasible solution than physician redistribution. Additionally, examining how

telehealth affects the geographic tie gradient—whether virtual care weakens the importance of physical proximity to family and training networks—will be crucial as technology reshapes healthcare delivery.

The persistence of geographic distribution concerns despite decades of policy intervention reflects the deep importance of preference and individual heterogeneity. Policies that acknowledge rather than attempt to overcome these attributes offer the most promising path toward positive and effective influence per dollar. This requires acknowledging that solutions to having more physicians, irrespective of the theoretical root causes, in some areas, may have to do with investments long before medical school, in the communities that shape the preferences physicians carry throughout their careers.

13 Appendix

13.0.1 Foreign Medical Graduates, and Foreign-Born Physicians

Table 6 presents the differences in location choices between foreign and U.S. medical graduates. Foreign medical graduates comprise around 25 percent of the sample, but only 11.2 percent of those returning to practice in their birth state, compared to 35.4 percent among those choosing other states ($t=51.71$). This 24-percentage-point gap mechanically reflects that many foreign medical graduates were born outside the United States and thus lack U.S. birth state ties to influence their location decisions.

In terms of rural practice, as shown in Table 13, only 4.9 percent of foreign-born physicians, comprising approximately 12 percent of sample, choose rural practice compared to 6.0 percent of U.S.-born physicians ($t=3.59$), a 18 percent lower rural practice rate. The overlap between foreign birth and foreign medical training is substantial, 89.8 percent of foreign-born physicians attended foreign medical schools, while 34.8 percent of foreign medical graduates were foreign-born. This partial overlap suggests that foreign birth and foreign medical training represent distinct but related barriers to rural practice. In the sample, foreign-born physicians are also older (35.9 vs 34.4 years) and less likely to be female (44.5 percent vs 48.6 percent), potentially reflecting immigration and visa processes that delay entry into U.S. practice.

[Table 13 about here.]

14 Technical Notes

14.1 Handling of AMA Data

The AMA delivers three slightly different codebooks (2016; 2017; 2018–2023). I harmonise variable names across layouts, merge residency files to recover specialty and training sites, and collapse 179 subspecialties to the 20 AMA major specialties. To match the wage data

described below, those 20 groups are further mapped into the seven Standard Occupational Classification (SOC) codes that the Bureau of Labor Statistics (BLS) used through 2018. Although the BLS adopted a new SOC system in late 2018, the 2010 definitions cover nearly the entire sample window and are therefore retained.⁴

Where residency and fellowship specialties differ, the most recent training defines a physician’s specialty; for the 4.7 % of cases reporting only a self-declared fellowship title, I back-map that title to the most common residency field implied by the fellowship.⁵

One challenge in processing AMA data is processing typed input. I use text processing to identify birthplace from physician’s input directly from Masterfile profiles, allowing a clean foreign-born indicator that is free of the misclassification problems in IMG status. Foreign medical school graduates physicians account for 31 % and 27 % of urban and rural shares in observed location, respectively.

14.2 Choice qualifier

To decide if a location choice, for example, Rural New York, is in the choice set for Pediatricians, I use BLS data aggregated to the choice level across the study period, and the choices of Pediatricians in the sample. Starting with a full choice set, I then omit choices that do not pass 2 criteria for *employment feasibility*, either: 1. have never been chosen by any physicians in that specialty across 4 years and 2. Have never had salaries reported by the BLS for existing physicians residing in that choice. For choices that do not fall into these two “feasibility criteria”, I impute using time-series data for that choice (in the case some years are missing for a choice present else where) or the closest similar choice, combined with time, state and specialty fixed effects. A robustness test where this choice set formation is tested against a full set of all US choices or a set of only GME present choices show that results varies by 10 percent when the choice set is not specialty specific and practically unchanged

⁴The full crosswalk appears in Appendix B along with checks that the handful of 2019 cases falling under the new SOC scheme do not drive results.

⁵For example, “Allergist” is assigned to Internal Medicine because that residency path is more prevalent than Pediatrics over 2016–2019. Dropping these physicians leaves the coefficients essentially unchanged.

if the choices with no GME funding are excluded.

14.3 OEWS Data

The Occupational Employment and Wage Statistics Model-Based 3-year estimates (OEWS-MB3) surveys approximately 1.2 million establishments over a rolling 3-year cycle and covers over 80 million workers. The data cover employer-reported wages across 12 intervals and use modeling to impute wages for non-responding establishments. OEWS captures base pay, tips, commissions, and cost-of-living allowances but excludes overtime pay, irregular bonuses, and shift differentials— compensation components particularly relevant for physicians who often work extended hours. Additionally, OEWS assumes standard full-time hours (2,080 hours annually) and reports occupational averages, whereas actual physician compensation varies widely based on individual work patterns, call schedules, and practice settings. Despite these limitations, OEWS remains a comprehensive source for location-specific wage data.

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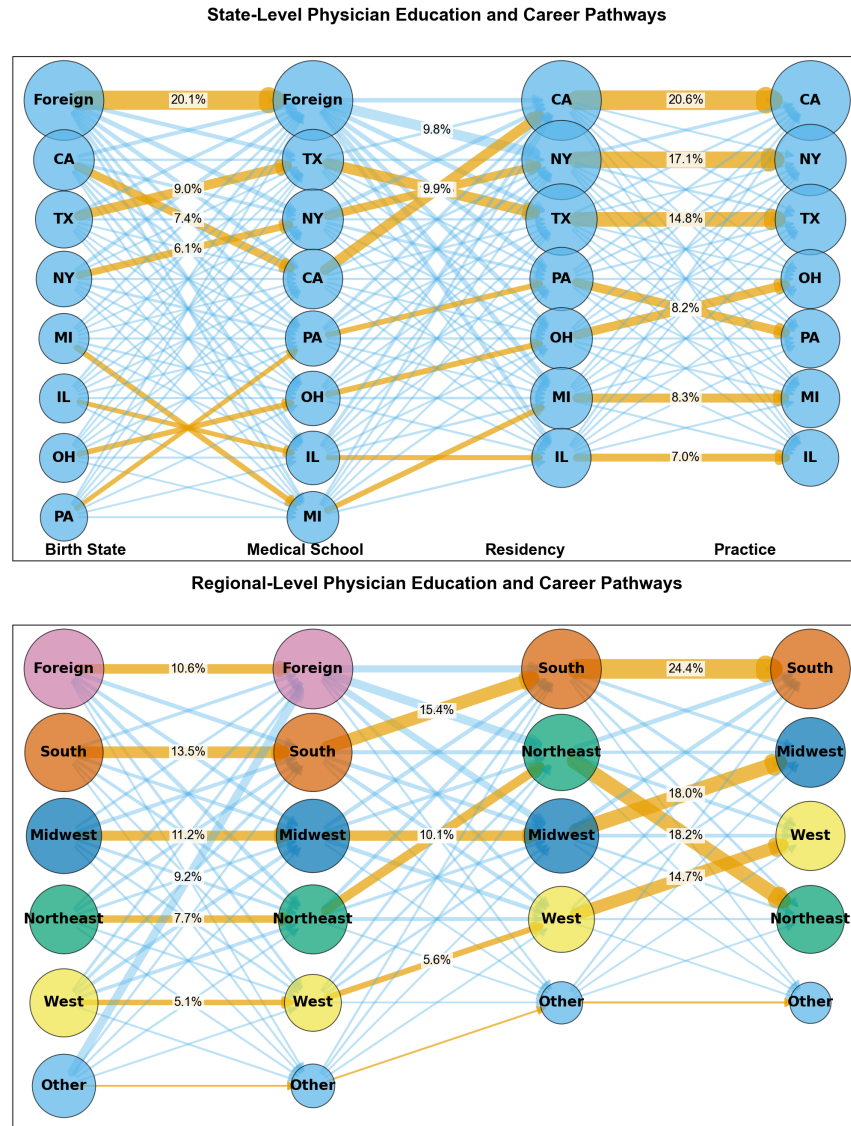
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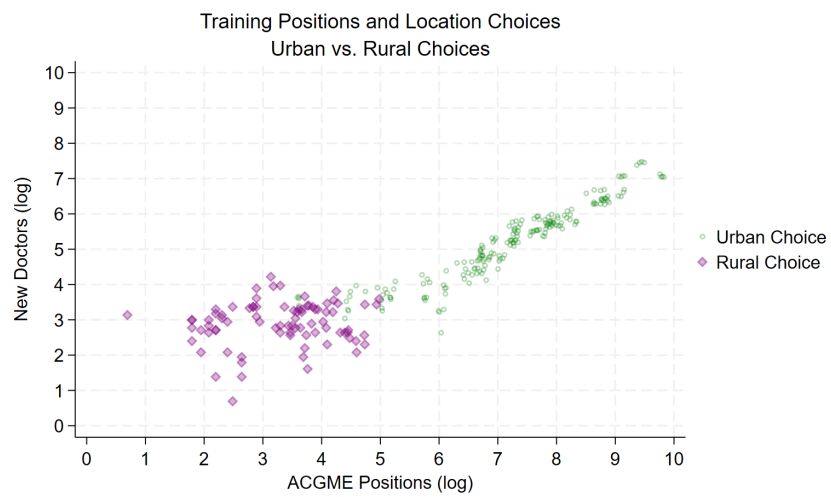
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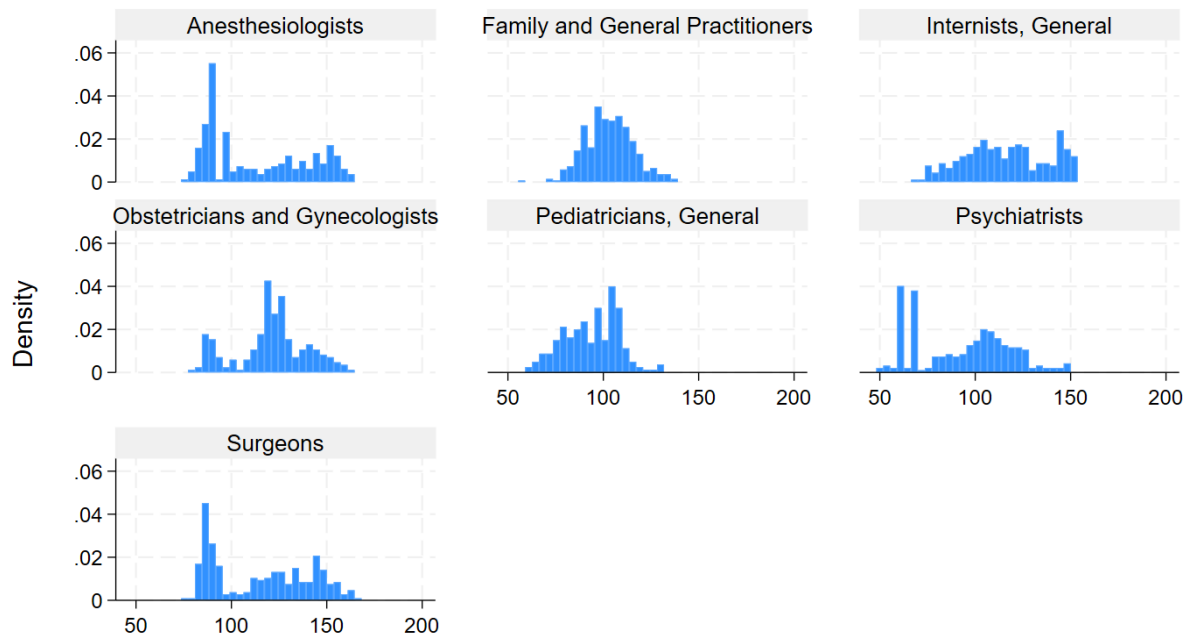
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Source: Author's calculations using AMA Masterfile data (2016-2019)

Figure 1: Persistence in Location Choice: Flows at State vs Region Level
 Note: Yellow flows are same-state, largest flow from each node. Only top choices are shown.





Average Hourly Wages, by Specialty (2015-2018)

Wages are measured in 2015 dollar and data are collected at the state-rural level, then calculated by specialty where data are available.

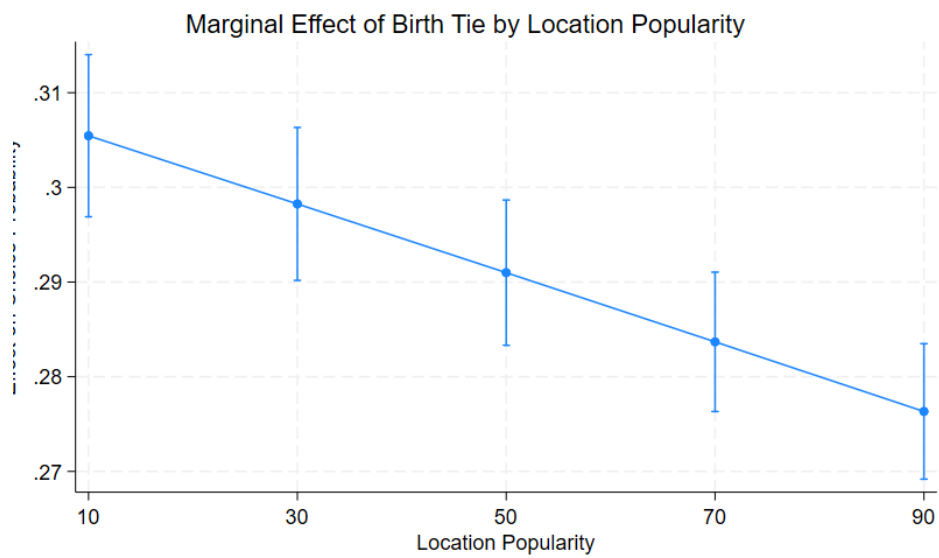


Figure 2: Marginal Effects of Birth State Match

Rural / Small-Metro Program Hub Hypothesis - Primary Care Specialties

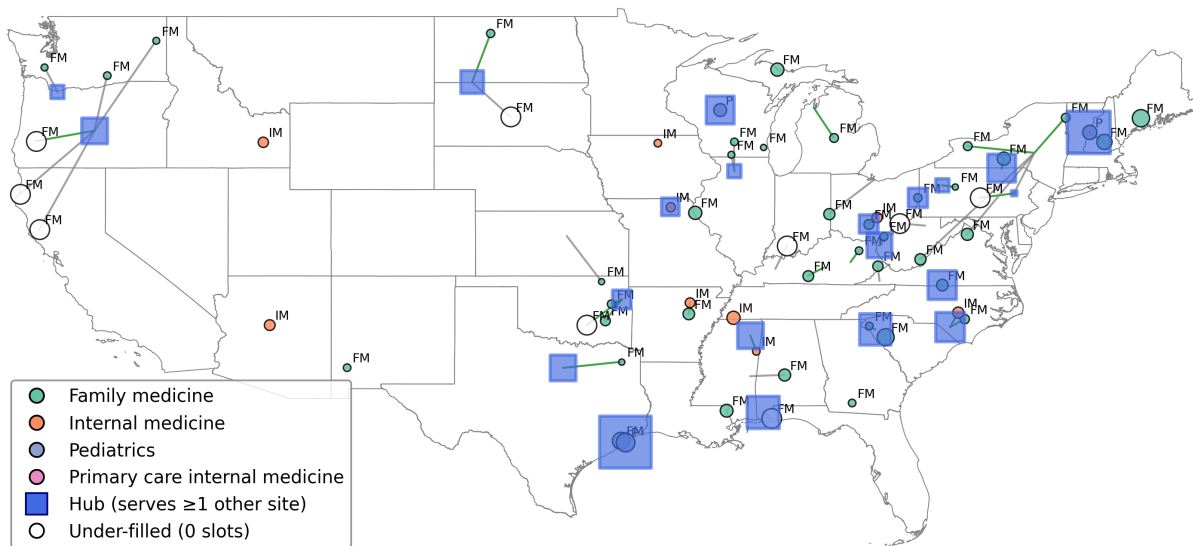


Figure 3: Toy example of selective GME hubs

Table 1: Summary Statistics of Full Sample

Variable	Full Sample		Urban Practice		Rural Practice	
	Mean	SD	Mean	SD	Mean	SD
Panel A: Physician Characteristics						
Age	34.61	4.20	34.62	4.17	34.54	4.62
Female	0.48	0.50	0.48	0.50	0.46	0.50
MD degree (vs DO)	0.92	0.27	0.92	0.27	0.86	0.35
Foreign medical graduate (IMGs)	0.31	0.46	0.31	0.46	0.27	0.44
Foreign born	0.12	0.32	0.12	0.32	0.10	0.30
Panel B: Training History						
Rural residency training	0.01	0.10	0.01	0.08	0.08	0.27
Rural medical school	0.03	0.17	0.02	0.14	0.11	0.31
Rural birth state	0.10	0.30	0.08	0.27	0.28	0.45
Chose birth state	0.17	0.37	0.19	0.39	0.26	0.44
Chose med school state	0.38	0.49	0.38	0.49	0.42	0.49
Chose residency state	0.62	0.49	0.63	0.48	0.57	0.50
Observations	58,647		55,174		3,473	

Notes: Data from AMA Masterfile 2016–2019. Rural defined using Rural–Urban Continuum Codes. Match variables indicate physician chose practice location in same location as birth, medical school, or residency. For residency, since the impact of specific location has to be separated from rural training exposure, the location match is at the state-rurality level, while the birth and medical school matches are at the state level.

Table 2: Physician Distribution and Rural Practice Rates by Specialty

Specialty	Sample Size	Share of Sample (%)	Rural Rate (%)	Avg Age	Female (%)
Anesthesiology	4,062	6.9	3.7	34.2	34.5
Family Medicine	11,690	19.9	12.1	34.1	49.9
Internal Medicine	19,811	33.8	4.1	34.8	41.9
Obstetrics/Gynecology	3,412	5.8	7.0	33.4	82.1
Pediatrics	7,766	13.2	3.8	33.9	69.6
Psychiatry	5,344	9.1	3.1	35.9	50.5
Surgery	6,562	11.2	6.0	35.6	26.6
Total	58,647	100.0	5.9	34.6	48.0

Notes: Specialties correspond to SOC system in BLS salary data.

Table 3: Choice Set Size by Specialty and Year

Specialty	Year			
	2016	2017	2018	2019
Anesthesiologists	64	66	67	70
Family Medicine	95	95	95	94
Internists, General	90	93	94	92
Obstetricians and Gynecologists	79	81	78	73
Pediatricians, General	85	81	79	84
Psychiatrists	67	78	73	73
Surgeons	85	83	83	82
<i>Column total</i>	565	577	569	568

Notes: Choices are assumed to start at full set, then omitted by criteria discussed in Technical Notes to be specialty-specific. Robustness tests include a test on this choice set definition.

Table 4: Summary Statistics of Choices by Rural–Urban Status

Variable	Full		Urban		Rural	
	Mean	SD	Mean	SD	Mean	SD
Price	112.45	23.01	111.63	22.93	113.69	23.08
Median income	57,788.12	11,383.08	63,237.78	10,133.40	49,619.61	7,659.83
Minimum temperature (F)	43.51	8.58	44.66	8.37	41.79	8.61
Black population	519,650.67	835,672.24	806,731.74	971,727.13	89,343.86	146,135.56
Primary Care GME FTEs	642.30	1,101.73	928.95	1,237.22	35.73	42.88
Observations	2,279		1,367		912	

Notes: Summary statistics calculated from analytic choice set (2016–2019). GME infrastructure data are not available for some choices, where they will be assumed to be 0.

Table 5: Summary Statistics of Graduate Medical Education Variables, Pooled 2016-2019

Variable	Urban Areas Mean (SD)	Rural Areas Mean (SD)	Urban/Rural Ratio
GME Funding (\$1000s)			
Total GME	271,908 (429,762)	4,351 (7,425)	62.5
Primary Care FTE	927 (1,242)	20.6 (35.0)	45.0
Resident Caps			
Total DME Resident Cap	2,075 (3,836)	31.0 (77.3)	67.0
Standard IME Resident Cap	1,711 (2,548)	28.6 (74.0)	59.8
Hospital Infrastructure			
Number of Beds	7,950 (8,858)	276 (356)	28.8
Positions per Unit	2,748 (3,510)	15.6 (28.4)	176.2
Programs per Unit	228 (269)	1.4 (2.5)	162.9
Distribution			
% with Zero GME	0.0	16.0	—
% with > 10 Programs	89.3	3.7	—
% with > 100 Positions	86.2	3.2	—
State-Years	196	187	

Notes: Data from Hospital Cost Reports (HCRIS) and ACGME are pooled across 2016-2019. Rurality are determined by RUCA. Standard deviations in parentheses. Standardized variables have mean 0 and SD 1 in the full sample. Primary Care Share calculated as Primary Care FTE divided by total resident positions. Ratio column shows urban mean divided by rural mean.

Table 6: Birthplace Impact on Practice Descriptives

	Return to Birth State	Other States	Difference (t-stat)
Panel A: Physician Characteristics			
Sample Size	11,625	47,021	
Share of Sample (%)	19.8	80.2	
Age (years)	33.65	34.85	-1.20*** (-27.69)
Female (%)	50.28	47.57	2.71*** (5.24)
Foreign Medical Graduate (%)	11.22	35.35	-24.13*** (-51.71)
Panel B: Geographic Flow Patterns			
	Share Staying in Same State (%)		
Birth State → Medical School		42.3	
Medical School → Residency		38.7	
Residency → Practice		62.0	
Birth State → Practice (Direct)		19.8	
Panel C: Correlation			
Correlation (birth ties, location chosen)		0.615	
Marginal effect at 10th percentile location popularity		0.305	
Marginal effect at 90th percentile location popularity		0.276	
Reduction in effect (%)		9.5	

Notes: Panel A compares physicians returning to birth states versus those choosing other states. Panel B shows persistence rates across career stages. Panel C demonstrates how birth state effects vary with location popularity, also shown in 2. *** $p < 0.01$.

Table 7: Conditional Logit Model of Location Choice (Main Analysis)

Variable	Coefficient (1)	Std. Error (2)	Odds Ratio (3)
<i>Location Characteristics</i>			
Rural	−0.735***	(0.029)	0.479
Median HH Income (std)	0.164***	(0.008)	1.178
Log Compensation (std)	0.012*	(0.006)	1.012
Mean Min. Temperature (std)	0.232***	(0.006)	1.261
% Black Population (std)	−0.012	(0.007)	0.988
Primary Care Share (std)	−0.023**	(0.011)	0.977
<i>GME Program Characteristics</i>			
Programs per Unit (std)	1.827***	(0.042)	6.213
Positions per Unit (std)	−1.551***	(0.040)	0.212
<i>Geographic Ties</i>			
Birth State	1.447***	(0.021)	4.249
Medical School State	1.688***	(0.019)	5.410
Residency State & Rurality	3.891***	(0.010)	48.958
<i>Rural Interactions</i>			
Rural Residency × Rural	0.207**	(0.095)	1.230
Foreign Med School × Rural	−0.486***	(0.040)	0.615
Log Compensation (std) × Rural	−0.015	(0.019)	0.986
Young Age × Rural	0.287***	(0.043)	1.333
Observations	N = 5,043,461		

Notes: Odds ratios in column (3) calculated as $\exp(\beta)$. Variables denoted with (std) are standardized; coefficients represent the effect of a one standard deviation change.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

Table 8: Policy Counterfactual Analysis: Impact on Rural Physician Supply

Policy Scenario	New Probability (1)	Change (pp) (2)	Change (%) (3)
<i>Panel A: GME Infrastructure Interventions (Rural Locations Only)</i>			
Add 10 programs to all rural hospitals	[-]	[-]	[-]
Provide 20 programs to zero-program hospitals	[-]	[-]	[-]
Set 5-program minimum for all hospitals	[-]	[-]	[-]
Expand existing programs by 50%	[-]	[-]	[-]
Double all rural programs	[-]	[-]	[-]
Double all rural positions	[-]	[-]	[-]
<i>Panel B: Physician Pipeline Interventions</i>			
Increase rural-trained physicians by 10pp ^a	[-]	[-]	[-]
Increase rural-trained physicians by 25pp	[-]	[-]	[-]
Increase rural-origin physicians by 10pp ^b	[-]	[-]	[-]
Increase rural-origin physicians by 25pp	[-]	[-]	[-]
<i>Panel C: Financial Incentives</i>			
Increase rural compensation by 20%	[-]	[-]	[-]
Increase rural compensation by 35%	[-]	[-]	[-]
Increase rural compensation by 50%	[-]	[-]	[-]
<i>Panel D: Choice Set Restrictions</i>			
Remove home location options ^c	[-]	[-]	[-]
<i>Panel E: Targeted Expansion Programs</i>			
Add 1,000 positions over 4 years (40% rural-targeted) ^d	[-]	[-]	[-]
Add 500 positions over 2 years (40% rural-targeted)	[-]	[-]	[-]

Notes: Counterfactual simulations based on conditional logit estimates from Table 7. Baseline rural choice probability is 0.0595 (N=58,647). Changes in column (2) are percentage points; column (3) shows percent change relative to baseline. Delta-method standard errors in brackets.

^a Assigns rural training status to additional physicians, distributed across 20 high-rurality states.

^b Modifies birth state origins to rural states for randomly selected physicians.

^c Excludes birth, medical school, and residency states from choice set (minimum 10 alternatives required).

^d New positions assigned characteristics favoring rural practice (rural training and origin).

Rural defined using Rural-Urban Continuum Codes. Primary care includes family medicine, internal medicine, and pediatrics.

Table 9: Policy Counterfactual Analysis: Impact on Rural Physician Supply

Policy Scenario	New Probability (1)	Change (pp) (2)	Change (%) (3)
<i>Panel A: GME Infrastructure Interventions (Rural Locations Only)</i>			
Add 10 programs to all rural hospitals	0.480	1.53	3.3
Provide 20 programs to zero-program hospitals	0.482	1.66	3.6
Set 5-program minimum for all hospitals	0.470	0.53	1.1
<i>Panel B: Rural Training Interventions</i>			
Convert urban-trained to rural-trained to add 10% ^a	0.463	-0.23	-0.5
Convert urban-trained to rural-trained to add 25%	0.457	-0.76	-1.6
Convert urban-born to rural-born to add 10% ^b	0.465	0.03	0.1
Convert urban-born to rural-born to add 25%	0.466	0.05	0.1
<i>Panel D: Choice Set Restrictions</i>			
Remove home location options ^c	0.544	7.94	17.1
<i>Panel E: Targeted Expansion Programs</i>			
Add 1,000 positions over 4 years (40% rural-targeted) ^d	0.492	2.74	5.9
Add 500 positions over 2 years (40% rural-targeted)	0.479	1.44	3.1

Notes: Counterfactual simulations based on conditional logit estimates from Table 7. Baseline rural choice probability is 0.465 (N=170 rural-trained physicians). Changes in column (2) are percentage points; column (3) shows percent change relative to baseline. Delta-method standard errors in brackets.

^a Assigns rural training status to additional physicians, distributed across 20 high-rurality states.

^b Modifies birth state origins to rural states for randomly selected physicians.

^c Excludes birth, medical school, and residency states from choice set (minimum 10 alternatives required).

^d New positions assigned characteristics favoring rural practice (rural training and origin).

Rural defined using Rural-Urban Continuum Codes. Primary care includes family medicine, internal medicine, and pediatrics.

Table 10: Cost-Effectiveness of Alternative Policy Interventions

Policy Intervention	Change in Rural Probability (pp)	Estimated Cost (\$ millions)	Cost per Additional Rural Physician
Infrastructure Investments			
Add 10 GME programs (rural)	[-]	[-]	[-]
Double rural GME positions	[-]	[-]	[-]
Pipeline Interventions			
Increase rural medical school class 25%	[-]	[-]	[-]
Expand rural residencies 25%	[-]	[-]	[-]
50% Increase from Current Rural Physicians	[-]	[-]	[-]
Financial Incentives			
20% rural compensation increase	[-]	[-]	[-]
Targeted Programs			
1,000 new positions (40% rural-targeted)	[-]	[-]	[-]
Near Rural Programs	[-]	[-]	[-]

Table 11: Residency Match Coefficient by Selection, Specialty, Time Periods and Birth Origins

Test	Residency Coefficient	% Change from Main Specification
Selection Tests:		
Foreign Born (most constrained)	3.997	+3.6%
FMGs (constrained)	4.030	+4.5%
US Born (less constrained)	3.837	-0.5%
US Medical Graduates (least constrained)	3.760	-2.5%
Specialty Competition:		
Non-surgical Specialties (less competitive)	3.947	+2.3%
Surgical Specialties (more competitive)	3.182	-17.5%
Time Periods:		
2016-2017	3.830	-0.7%
2018-2019	3.883	+0.6%

Notes: Selection Tests use variation in the dummy for residency match among categories of Foreign Born, FMGs (physicians with foreign medical school degrees), US Born, and USMG (US physicians with US medical school degrees).

Table 12: Robustness Tests of Model Specifications using the Coefficient on Rurality

Specification	Rural Coefficient	Residency Match	Birth Match	GME Programs per Unit	Pseudo R^2
Panel A: Sequential Controls					
Rural only	-2.458***	—	—	—	0.077
Add GME infrastructure	-1.489***	—	—	2.244***	0.159
Add economic controls	-1.477***	—	—	2.247***	0.159
Add demographics	-1.268***	—	—	2.066***	0.163
Add geographic ties (Main)	-1.627***	3.891***	1.447***	1.797***	0.532
Panel B: Choice Set Definitions					
Baseline (historically chosen)	-1.633***	3.858***	1.473***	1.812***	0.533
Expanded (all possible)	-1.881***	4.213***	1.535***	1.798***	0.528
Panel C: Sample Restrictions					
Full sample	-1.627***	3.858***	1.458***	1.797***	0.532
25% random sample	-1.614***	3.871***	1.462***	1.801***	0.529

Notes: Standard errors clustered by physician. *** p<0.01.

Table 13: Foreign Birth Status and Physician Location Choice

	Foreign Birth Status		
	US-Born (N=51,695)	Foreign-Born (N=6,952)	Difference (t-stat)
Panel A: Rural Practice Rate			
Choose Rural Practice (%)	6.03 (0.238)	4.95 (0.217)	-1.08*** (3.59)
Panel B: Physician Characteristics			
Age (years)	34.44 (4.10)	35.88 (4.70)	1.44*** (27.03)
Female (%)	48.59 (0.500)	44.48 (0.497)	-4.11*** (6.45)
Foreign Medical Graduate (%)	22.60 (0.418)	89.77 (0.303)	67.17*** (126.84)
Panel C: Foreign Birth-Medical Training Overlap			
<i>Among Foreign-Born Physicians:</i>			
Foreign Medical School (%)		89.77	
US Medical School (%)		10.23	
<i>Among Foreign Medical Graduates:</i>			
Foreign-Born (%)		34.82	
US-Born (%)		65.18	

Notes: Standard deviations in parentheses for means. t-statistics in parentheses for differences. Foreign-born physicians comprise 11.85% of the sample.

*** p<0.01, ** p<0.05, * p<0.10